

Formula Sheet:

1. Discrete compounding: $B_n = B_0(1 + r)^n$.
2. Continuous compounding: $B_t = B_0 e^{rt}$.
3. Present Value of a bond under discrete compounding interest rate:

$$PV_B = C \frac{1}{r} \left(1 - \frac{1}{(1 + r)^T} \right) + \frac{F}{(1 + r)^T},$$

C : coupon, T : time to maturity, F : face.

4. Payoff of European Call with strike K and maturity T : $(S_T - K)^+$.
5. Payoff of European Put with strike K and maturity T : $(K - S_T)^+$.
6. Fair delivery price of a forward contract: $K(t, T) = S_t e^{r(T-t)}$.
7. Law of One Price: In an arbitrage-free market, two portfolios (or assets or cashflows) with identical future values must have identical present values.
8. Put-Call Parity: $c_t + K e^{-r(T-t)} = p_t + S_t$, where c_t is premium of the call at time t , and p_t is premium of the put at time t .
9. Value process in multi-period model: $V_n^h = x_n B_n + y_n S_n = x_n (1 + r)^n + y_n S_n$.
10. A portfolio strategy $h_n = (x_n, y_n)$ is said to be a self-financing strategy if

$$x_n B_n + y_n S_n = x_{n+1} B_n + y_{n+1} S_n$$

for all $n = 1, \dots, N - 1$.

11. The multi-period model is free of arbitrage if and only if $d < 1 + r < u$ (discrete comp.).
12. Risk-neutral probabilities in multi-period models (discrete comp.):

$$S_n = \frac{1}{1 + r} \mathbb{E}^{\mathbb{Q}}[S_{n+1}], \quad q_u = \frac{(1 + r) - d}{u - d}, \quad q_d = \frac{u - (1 + r)}{u - d}.$$

13. Replicating portfolio for the claim whose contract function is Φ in 1-period model (discrete comp.):

$$x = \frac{1}{1 + r} \frac{u\Phi(d) - d\Phi(u)}{u - d}, \quad y = \frac{1}{S_0} \frac{\Phi(u) - \Phi(d)}{u - d}.$$

14. Risk-neutral pricing for European type claim $X = \Phi(S_N)$ in N -period market (discrete comp.):

$$\Pi_0(X) = \left(\frac{1}{1 + r} \right)^N \sum_{k=0}^N \binom{N}{k} q_u^k q_d^{N-k} \Phi(S_0 u^k d^{N-k}).$$

15. Risk-neutral pricing for American type claim $X = \Phi(S_N)$ in N -period market (discrete comp.):

$$\Pi_n(X)|_{S_n=s} = \max \left\{ \Phi(s), \frac{1}{1 + r} \left(q_u \Pi_{n+1}(X)|_{S_{n+1}=us} + q_d \Pi_{n+1}(X)|_{S_{n+1}=ds} \right) \right\}.$$

16. A stochastic process $W = (W_t)_{t \geq 0}$ is called a Brownian motion if it satisfies:

- (a) $\mathbb{P}(W_0 = 0) = 1$.
- (b) For all $0 \leq s < t$, the random variable $W_t - W_s \sim N(0, t - s)$.
- (c) W has independent increments, i.e. for any $r < s \leq t < u$ then $W_u - W_t$ and $W_s - W_r$ are independent random variables.
- (d) Almost all sample paths are continuous.

17. Normal distribution $N(\mu, \sigma^2)$ has probability density function $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}$.

18. Let (X, Y) be bivariate normal and $\mathbb{E}[X] = \mathbb{E}[Y] = 0$, then X and Y are independent iff $\mathbb{E}[XY] = 0$.

19. Itô isometry:

$$\mathbb{E} \left[\left(\int_0^T X_s dW_s \right)^2 \right] = \mathbb{E} \left[\int_0^T |X_s|^2 ds \right], \quad \mathbb{E} \left[\left(\int_0^T X_s dW_s \right) \left(\int_0^T Y_s dW_s \right) \right] = \mathbb{E} \left[\int_0^T X_s Y_s ds \right].$$

20. Properties of Itô integrals:

- (a) $\int_S^T g_t dW_t = \int_S^U g_t dW_t + \int_U^T g_t dW_t$,
- (b) $\int_0^T (\alpha g_t + f_t) dW_t = \alpha \int_0^T g_t dW_t + \int_0^T f_t dW_t$,
- (c) $\mathbb{E} \left[\int_0^t g_s dW_s \right] = 0$.

21. Itô's formula: for $Y_t := f(t, X_t)$,

$$\begin{aligned} dY_t &= \frac{\partial f}{\partial t}(t, X_t) dt + \frac{\partial f}{\partial x}(t, X_t) dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(t, X_t) \langle dX_t, dX_t \rangle \\ &= \left[\frac{\partial f}{\partial t}(t, X_t) + U_t \frac{\partial f}{\partial x}(t, X_t) + V_t^2 \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(t, X_t) \right] dt + \left[V_t \frac{\partial f}{\partial x}(t, X_t) \right] dW_t, \end{aligned}$$

where $dX_t = U_t dt + V_t dW_t$.

22. $\langle dt, dt \rangle = \langle dt, dW_t \rangle = \langle dW_t, dt \rangle = 0$, $\langle dW_t, dW_t \rangle = dt$.

23. Geometric Brownian motion:

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad S_t = S_0 \exp \left(\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right), \quad \mathbb{E}(S_t) = S_0 e^{\mu t}$$

24. Black-Scholes PDE:

$$\frac{\partial f}{\partial t}(t, x) + rx \frac{\partial f}{\partial x}(t, x) + \frac{1}{2} \sigma^2 x^2 \frac{\partial^2 f}{\partial x^2}(t, x) - rf(t, x) = 0$$

with the terminal condition

$$f(T, x) = \Phi(x),$$

where Φ is the payoff function of the option, $\sigma > 0$ is the volatility and r is the interest rate.

25. Feynman-Kac formula: Assume that the function $f : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ is a solution to the following terminal value problem PDE

$$\frac{\partial f}{\partial t}(t, x) + \mu(t, x) \frac{\partial f}{\partial x}(t, x) + \frac{1}{2} \sigma^2(t, x) \frac{\partial^2 f}{\partial x^2}(t, x) - rf(t, x) = 0, \quad (1)$$

$$f(T, x) = \Phi(x). \quad (2)$$

For any $(t, x) \in [0, T] \times \mathbb{R}$, define the stochastic process $(X_s)_{s \in [t, T]}$ as the solution to the SDE

$$dX_s = \mu(s, X_s)ds + \sigma(s, X_s)dW_s, \quad \forall s \in [t, T], \quad X_t = x.$$

Assume that the stochastic process $(e^{-rs}\sigma(s, X_s)\frac{\partial f}{\partial x}(s, X_s))_{s \in [t, T]} \in L^2([0, T] \times \mathbb{R})$.

Then the solution f of (1)-(2) can be expressed as

$$f(t, x) = e^{-r(T-t)}\mathbb{E}[\Phi(X_T)|X_t = x] \quad \forall (t, x) \in [0, T] \times \mathbb{R}.$$

CHECKING

1. Consider a two-period binomial model, in which a non-dividend-paying stock $S = (S_t)_{t=0,1,2}$ has its current price of $S_0 = £120$. Over the next two 4-month periods, the stock price is expected to either increase by 10% or decrease by 6%. Let us assume that the risk-free rate of interest is 6% per annum (continuous compounding).
- Show that the market is free of arbitrage. Then find the risk-neutral probabilities. Please round your solution to 1 decimal place. [6 marks]
 - Suppose there is an 8-month European call option written on S with strike price $K = £124$.
 - What is the arbitrage-free price of this option at $t = 0$? Please round your answer to 3 decimal places. [7 marks]
 - Suppose Amy bought this option at $t = 0$, then decided to sell this contract after 4 months. How much should Amy sell it for after 4 months (i.e. at $t = 1$) so that there is no arbitrage opportunity? Please round your solution to 3 decimal places. [6 marks]
 - Suppose Katie short sells this call option immediately at $t = 0$. To hedge her short position in this option, Katie adopts a delta-hedging strategy by taking a long position in S . Assuming she can hold a fractional number of stock units, how many units of stock must she buy to achieve a perfect hedge? Round your answer to 3 decimal places. By showing the portfolio is riskless (round the answer to two decimal places), verify that it results in a perfect hedge. [6 marks]
 - Find the arbitrage-free price of an 8-month American put option written on S with strike price $K = £124$. Please round your answer to 3 decimal places. Is an early exercise optimal? [13 marks]

[Total for this question: 38 marks]

Solution

- (a) We have $u = 1.1$, $d = 0.94$, $r = 6\%$ and $\Delta t = \frac{1}{3}$. Since

$$d < e^{r\Delta t} \approx 1.02 < u,$$

we conclude that the market is arbitrage-free. [4 marks]

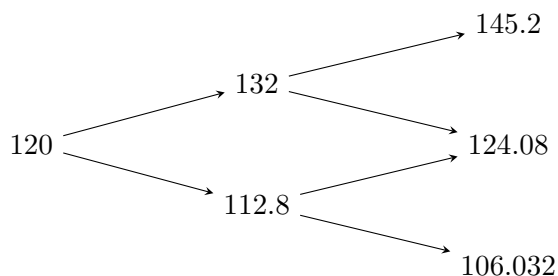
Applying the formulae, we may find

$$q_u = \frac{e^{r\Delta t} - d}{u - d} = 0.5,$$

$$q_d = 1 - q_u = 0.5.$$

[2 marks]

- (b) i. The evolution of the stock price is shown as in the figure below.



Therefore, as the payoff of the call option is

$$\Phi(S_2) = (S_2 - K)^+ = (S_2 - 124)^+,$$

[1 mark]

we deduce that

$$\Phi(S_2(uu)) = (145.2 - 124)^+ = 21.2,$$

$$\Phi(S_2(ud)) = (124.08 - 124)^+ = 0.08,$$

$$\Phi(S_2(dd)) = (106.032 - 124)^+ = 0.$$

[3 marks]

Therefore, the risk-neutral valuation of the option is

$$\begin{aligned}\Pi_0(\Phi) &= e^{-2r\Delta t} \mathbb{E}[\Phi(S_2)] \\ &= e^{-0.04} \left(0.5^2 \cdot 21.2 + 2 \cdot 0.5^2 \cdot 0.08 + 0.5^2 \cdot 0 \right) \\ &\approx 5.131.\end{aligned}$$

[3 marks]

The price of the option is £5.131.

ii. If $S_1 = S_1(u) = 132$, then the arbitrage-free price is

$$\begin{aligned}\Pi_1(\Phi)|_{S_1=S_1(u)} &= e^{-r\Delta t} (q_u \Phi(S_2(uu)) + q_d \Phi(S_2(ud))) \\ &= e^{-0.02} \left(0.5 \cdot 21.2 + 0.5 \cdot 0.08 \right) \\ &\approx 10.429.\end{aligned}$$

[3 marks]

If $S_1 = S_1(d) = 112.8$, then the arbitrage-free price is

$$\begin{aligned}\Pi_1(\Phi)|_{S_1=S_1(d)} &= e^{-r\Delta t} (q_u \Phi(S_2(ud)) + q_d \Phi(S_2(dd))) \\ &= e^{-0.02} \left(0.5 \cdot 0.08 + 0.5 \cdot 0 \right) \\ &\approx 0.039.\end{aligned}$$

[3 marks]

iii. At $S_0 = 120$,

$$\begin{aligned}\Delta &= \frac{\Phi(S_1(u)) - \Phi(S_1(d))}{S_0(u - d)} \\ &= \frac{8 - 0}{132 - 112.8} \\ &\approx 0.417,\end{aligned}$$

so Amy should hold 0.417 unit of the stock to hedge the short position in the option in the first period. [1 mark]

To verify that this strategy consisting of a short call and long Δ units of stock is riskless, we see that

$$-\Phi(S_1(u)) + 0.417S_1(u) \approx 47.04 \approx -\Phi(S_1(d)) + 0.417S_1(d).$$

Therefore the portfolio is riskless.

[1 mark]

If $S_1 = S_1(u) = 132$, then

$$\begin{aligned}\Delta &= \frac{\Phi(S_2(uu)) - \Phi(S_2(ud))}{S_2(uu) - S_2(ud)} \\ &= \frac{21.2 - 0.08}{145.2 - 124.08} \\ &= 1,\end{aligned}$$

so Amy should hold 1 unit of the stock to hedge the short position in the option. [1 mark]

If $S_1 = S_1(d) = 112.8$, then

$$\begin{aligned}\Delta &= \frac{\Phi(S_1(ud)) - \Phi(S_2(dd))}{S_2(ud) - S_2(dd)} \\ &= \frac{0.08 - 0}{124.08 - 106.032} \\ &= 0.004,\end{aligned}$$

so Amy should hold 0.004 units of the stock to hedge the short position in the option. [1 mark]

This hedging strategy consists of a short option and a long position in Δ units of S . When $S_1 = S_1(u) = 132$, $\Delta = 1$, and

$$-\Phi(S_2(uu)) + S_2(uu) = K = 124 = -\Phi(S_2(ud)) + S_2(ud),$$

so this portfolio is riskless.

[1 mark]

Similarly, when $S_1 = S_1(d) = 112.88$, $\Delta = 0.004$, and

$$-\Phi(S_2(ud)) + 0.004S_2(ud) \approx 0.42 \approx -\Phi(S_2(dd)) + 0.004S_2(dd),$$

so this portfolio is riskless.

[1 mark]

(c) The payoff of put is given by

$$\Phi(S) = (K - S)^+ = (124 - S)^+.$$

[1 mark]

At $t = 2$,

$$\begin{aligned}\Phi(S_2(uu)) &= (124 - 145.2)^+ = 0, \\ \Phi(S_2(ud)) &= (124 - 124.08)^+ = 0, \\ \Phi(S_2(dd)) &= (124 - 106.032)^+ = 17.968.\end{aligned}$$

[3 marks]

At $S_1(u)$:

$$\Pi_1(\Phi)|_{S_1(u)} = \max\{(124 - 132)^+, 0\} = 0, \quad \text{no early exercise.}$$

[3 marks]

At $S_1(d)$:

$$\begin{aligned}\Pi_1(\Phi)|_{S_1(d)} &= \max\{(124 - 112.8)^+, e^{-0.02}(0.5 \cdot 0 + 0.5 \cdot 17.968)\} \\ &= \max\{11.2, 8.806\} \\ &= 11.2, \quad \text{early exercise.}\end{aligned}$$

[3 marks]

At $t = 0$:

$$\begin{aligned}\Pi_0(\Phi) &= \max\{(124 - 120)^+, e^{-0.02}(0.5 \cdot 0 + 0.5 \cdot 11.2)\} \\ &= \max\{4, 5.489\} \\ &= 5.489, \quad \text{no early exercise.}\end{aligned}$$

[3 marks]

Therefore, the price is £5.489 and an early exercise is optimal.

Comment

- (a) and (b)(i) are standard.
- (b)(ii) is from a workshop sheet.
- (b)(iii) is not very standard, but one-period case was covered in lecture.
- (c) is standard.

CHECKING

2. Let $T > 0$. Assume a Black-Scholes market, where the evolution of stock price $(S_t)_{t \in [0, T]}$ is given by a geometric Brownian motion with drift r and volatility σ , and the bank account pays risk-free interest continuously at annual rate r , so that

$$S_t = s_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right),$$

where $(W_t)_{t \geq 0}$ is a standard Brownian motion, and $s_0 > 0$ is a real number.

Let $n \in \mathbb{N}$ be a positive integer. Consider the *power call* option Φ whose payoff function is given by

$$\Phi(x) = (x^n - K)^+$$

and the *power put* option Φ whose payoff function is given by

$$\Phi(x) = (K - x^n)^+.$$

- (a) Verify that $(S_t)_{t \geq 0}$ satisfies the stochastic differential equation

$$\begin{aligned} dS_t &= rS_t dt + \sigma S_t dW_t, \\ S_0 &= s_0. \end{aligned}$$

Then find the stochastic differential equation satisfied by the n -th power of the stock price process, $(S_t^n)_{t \geq 0}$, which is also a geometric Brownian motion. [9 marks]

- (b) Find the Black-Scholes price (or arbitrage-free price) $\Pi_0(\text{Call}(T, K))$ of the power call option with strike K , maturity T . Write down your answer using the cumulative distribution function (CDF) $N(x)$ of the standard normal distribution $N(0, 1)$, $d_-(n)$ and $d_+(n)$, where

$$d_-(n) = \frac{n(r - \frac{\sigma^2}{2})T + \ln \left(\frac{s_0^n}{K} \right)}{n\sigma\sqrt{T}},$$

and

$$d_+(n) = d_-(n) + n\sigma\sqrt{T} = \frac{n(r - \frac{\sigma^2}{2} + n\sigma^2)T + \ln \left(\frac{s_0^n}{K} \right)}{n\sigma\sqrt{T}}.$$

Hint: For $K > 0$, find $\mathbb{P}(S_T^n > K)$ and $\mathbb{E}[S_T^n 1_{\{S_T^n > K\}}]$. [15 marks]

- (c) Find the Black-Scholes price (or arbitrage-free price) $\Pi_0(\text{Put}(T, K))$ of the power put option with strike K , maturity T . [8 marks]
- (d) Using the explicit expressions for $\Pi_0(\text{Call}(T, K))$ and $\Pi_0(\text{Put}(T, K))$ found above, write down the Put-Call parity relation for power options. [4 marks]

[Total for this question: 36 marks]

Solution

- (a) By Itô's formula with $f(t, x) = s_0 e^{(r - \frac{1}{2}\sigma^2)t + \sigma x}$,

$$\begin{aligned} dS_t &= df(t, W_t) = s_0 \sigma e^{(r - \frac{1}{2}\sigma^2)t + \sigma W_t} dW_t + \frac{1}{2} s_0 \sigma^2 e^{(r - \frac{1}{2}\sigma^2)t + \sigma W_t} dt + s_0 (r - \frac{1}{2}\sigma^2) e^{(r - \frac{1}{2}\sigma^2)t + \sigma W_t} dt \\ &= \sigma s_0 e^{(r - \frac{1}{2}\sigma^2)t + \sigma W_t} dW_t + r s_0 e^{(r - \frac{1}{2}\sigma^2)t + \sigma W_t} dt \\ &= \sigma S_t dW_t + r S_t dt, \end{aligned}$$

[3 marks]

and

$$S_0 = s_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) \cdot 0 + \sigma W_0 \right) = s_0.$$

[1 mark]

By Itô's formula with $f(t, x) = x^n$,

$$dS_t^n = nS_t^{n-1}dS_t + \frac{1}{2}n(n-1)S_t^{n-2}d\langle S \rangle_t,$$

[1 mark]

and

$$d\langle S \rangle_t = \sigma^2 S_t^2 dt,$$

[1 mark]

so we have

$$dS_t^n = n\sigma S_t^n dW_t + \left(nr + \frac{1}{2}n(n-1)\sigma^2 \right) S_t^n dt$$

[2 marks]

with initial condition $S_0^n = s_0^n$.

[1 mark]

(b) As $S_t^n = s_0^n e^{n(r - \frac{1}{2}\sigma^2)t + n\sigma W_t}$, for $K > 0$,

$$\begin{aligned} \mathbb{P}(S_T^n > K) &= \mathbb{P} \left(s_0^n \exp \left\{ n \left(r - \frac{\sigma^2}{2} \right) T + n\sigma W_T \right\} > K \right) \\ &= \mathbb{P} \left(\left(r - \frac{\sigma^2}{2} \right) T + \sigma W_T > \ln \left(\frac{K^{\frac{1}{n}}}{s_0} \right) \right) \\ &= \mathbb{P} \left(W_T > \frac{\frac{1}{n} \ln \left(\frac{K}{s_0^n} \right) - \left(r - \frac{\sigma^2}{2} \right) T}{\sigma} \right) \\ &= \mathbb{P} \left(\sqrt{T} N(0, 1) > \frac{\frac{1}{n} \ln \left(\frac{K}{s_0^n} \right) - \left(r - \frac{\sigma^2}{2} \right) T}{\sigma} \right) \\ &= \mathbb{P}(N(0, 1) > -d_-(n)) \\ &= N(d_-(n)), \end{aligned}$$

[5 marks]

where

$$d_-(n) = \frac{n \left(r - \frac{\sigma^2}{2} \right) T + \ln \left(\frac{s_0^n}{K} \right)}{n\sigma\sqrt{T}}.$$

Using the above observation,

$$\begin{aligned}
 \mathbb{E}[S_T^n 1_{\{S_T^n > K\}}] &= \mathbb{E}\left[s_0^n e^{n(r - \frac{\sigma^2}{2})T + n\sigma W_T} 1_{\{S_T^n > K\}}\right] \\
 &= \mathbb{E}\left[s_0^n \exp\left\{n\left(r - \frac{\sigma^2}{2}\right)T + n\sigma\sqrt{T}N(0, 1)\right\} 1_{\{N(0,1) > -d_-(n)\}}\right] \\
 &= s_0^n \int_{-d_-(n)}^{\infty} \exp\left\{n\left(r - \frac{\sigma^2}{2}\right)T + n\sigma\sqrt{T}x\right\} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\
 &= s_0^n e^{nrT + \frac{1}{2}n(n-1)\sigma^2 T} \int_{-d_-(n)}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{(x - n\sigma\sqrt{T})^2}{2}} dx \\
 &= s_0^n e^{nrT + \frac{1}{2}n(n-1)\sigma^2 T} \int_{-d_-(n) - n\sigma\sqrt{T}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\
 &= s_0^n e^{nrT + \frac{1}{2}n(n-1)\sigma^2 T} \mathbb{P}(N(0, 1) > -d_+(n)) \\
 &= s_0^n e^{nrT + \frac{1}{2}n(n-1)\sigma^2 T} N(d_+(n)),
 \end{aligned}$$

[6 marks]

where

$$d_+(n) = d_-(n) + n\sigma\sqrt{T} = \frac{n(r - \frac{\sigma^2}{2} + n\sigma^2)T + \ln\left(\frac{s_0^n}{K}\right)}{n\sigma\sqrt{T}}.$$

Therefore, the price of the call is

$$\begin{aligned}
 \Pi_0(\text{Call}(T, K)) &= e^{-rT} \mathbb{E}[(S_T^n - K)^+] \\
 &= e^{-rT} \mathbb{E}[(S_T^n - K) 1_{\{S_T^n > K\}}] \\
 &= e^{-rT} \left(\mathbb{E}[S_T^n 1_{\{S_T^n > K\}}] - K \mathbb{P}(S_T^n > K) \right) \\
 &= e^{-rT} \left(s_0^n e^{nrT + \frac{1}{2}n(n-1)\sigma^2 T} N(d_+(n)) - K N(d_-(n)) \right)
 \end{aligned}$$

[4 marks]

(c) Similarly,

$$\begin{aligned}
 \Pi_0(\text{Put}(T, K)) &= e^{-rT} \mathbb{E}[(K - S_T^n)^+] \\
 &= e^{-rT} \mathbb{E}[(K - S_T^n) 1_{\{S_T^n \leq K\}}] \\
 &= e^{-rT} \left(K \mathbb{P}(S_T^n \leq K) - \mathbb{E}[S_T^n 1_{\{S_T^n \leq K\}}] \right).
 \end{aligned}$$

[3 marks]

Notice that

$$\mathbb{P}(S_T^n \leq K) = 1 - \mathbb{P}(S_T^n > K) = 1 - N(d_-(n)),$$

[1 mark]

and as

$$\mathbb{E}[S_T^n] = s_0^n e^{nrT + \frac{1}{2}n(n-1)\sigma^2 T},$$

[2 marks]

we have that

$$\begin{aligned}\mathbb{E}[S_T^n 1_{\{S_T^n \leq K\}}] &= \mathbb{E}[S_T^n] - \mathbb{E}[S_T^n 1_{\{S_T^n > K\}}] \\ &= s_0^n e^{nrT + \frac{1}{2}n(n-1)\sigma^2 T} (1 - N(d_+(n))).\end{aligned}$$

[1 mark]

Therefore,

$$\begin{aligned}\Pi_0(\text{Put}(T, K)) &= e^{-rT} \left(K(1 - N(d_-(n))) - s_0^n e^{nrT + \frac{1}{2}n(n-1)\sigma^2 T} (1 - N(d_+(n))) \right) \\ &= e^{-rT} \left(KN(-d_-(n)) - s_0^n e^{nrT + \frac{1}{2}n(n-1)\sigma^2 T} N(-d_+(n)) \right).\end{aligned}$$

[1 mark]

(d) The put-call parity relation is given by

$$\Pi_0(\text{Call}(T, K)) + Ke^{-rT} = \Pi_0(\text{Put}(T, K)) + s_0^n e^{(n-1)rT + \frac{1}{2}n(n-1)\sigma^2 T}.$$

[4 marks]

Comment

- (a) This part is standard.
- (b) This part is similar to what was covered in class, but the computation is more complicated.
- (c) This part is similar to what was covered in class.
- (d) This part is similar to a homework problem.

3. (a) In the *Vasicek interest rate model*, the interest rate is a stochastic process $(R_t)_{t \geq 0}$ which evolves according to the following stochastic differential equation

$$dR_t = (b - aR_t)dt + \sigma dW_t, \quad R_0 = r,$$

where $(W_t)_{t \geq 0}$ is a standard Brownian motion, and $r, a, b, \sigma > 0$ are positive real numbers.

- i. By considering $X_t = (R_t - \frac{b}{a}(1 - e^{-at}))e^{at}$, show that $(R_t)_{t \geq 0}$ is given by

$$R_t = re^{-at} + \frac{b}{a}(1 - e^{-at}) + \sigma e^{-at} \int_0^t e^{as} dW_s, \quad \forall t \geq 0.$$

[6 marks]

- ii. From the above representation, you may assume that for every $t > 0$, R_t has a normal distribution. Find its expectation $\mathbb{E}[R_t]$ and variance $\text{Var}(R_t)$.

[8 marks]

- (b) Let $T, \sigma, r, \theta > 0$ be fixed constants. Use the Feynman-Kac representation formula to solve the following equation subject to a terminal condition:

$$\begin{aligned} \frac{\partial u}{\partial t}(t, x) + \frac{1}{2}\sigma^2 \frac{\partial^2 u}{\partial x^2}(t, x) &= ru(t, x), \quad 0 \leq t \leq T, \\ u(T, x) &= e^{\theta x}. \end{aligned}$$

[12 marks]

[Total for this question: 26 marks]

Solution

- (a) i. First we write $X_t = e^{at}R_t - \frac{b}{a}(e^{at} - 1)$. Apply Itô's formula to X and get:

$$\begin{aligned} dX_t &= e^{at}dR_t + ae^{at}R_tdt - be^{at}dt \\ &= e^{at}(b - aR_t)dt + \sigma e^{at}dW_t + ae^{at}R_tdt - be^{at}dt \\ &= \sigma e^{at}dW_t. \end{aligned}$$

[4 marks]

Consequently, by integrating over the time interval $[0, t]$ and $X_0 = r$, we get that for all $t \geq 0$,

$$X_t = r + \int_0^t \sigma e^{as} dW_s$$

[2 marks]

which leads to the required result.

- ii. Using (i),

$$\begin{aligned} \mathbb{E}[R_t] &= \mathbb{E}[re^{-at}] + \mathbb{E}\left[\frac{b}{a}(1 - e^{-at})\right] + \mathbb{E}\left[\sigma e^{-at} \int_0^t e^{as} dW_s\right] \\ &= re^{-at} + \frac{b}{a}(1 - e^{-at}). \end{aligned}$$

[2 marks]

Similarly, by Ito isometry,

$$\begin{aligned}\mathbb{E}[R_t^2] &= \left(re^{-at} + \frac{b}{a}(1 - e^{-at})\right)^2 + 2\left(re^{-at} + \frac{b}{a}(1 - e^{-at})\right)\mathbb{E}\left[\sigma e^{-at} \int_0^t e^{as} dW_s\right] \\ &\quad + \mathbb{E}\left[\left(\sigma e^{-at} \int_0^t e^{as} dW_s\right)^2\right] \\ &= \left(re^{-at} + \frac{b}{a}(1 - e^{-at})\right)^2 + \sigma^2 e^{-2at} \int_0^t e^{2as} ds \\ &= \left(re^{-at} + \frac{b}{a}(1 - e^{-at})\right)^2 + \frac{\sigma^2}{2a}(1 - e^{-2at}).\end{aligned}$$

[4 marks]

Therefore, we have

$$\text{Var}(R_t) = \mathbb{E}[R_t^2] - \mathbb{E}[R_t]^2 = \frac{\sigma^2}{2a}(1 - e^{-2at}).$$

[2 marks]

- (b) Based on the PDE, we have $\mu(t, x) = 0$, $\sigma(t, x) = \sigma$.

[2 marks]

Therefore, the corresponding SDE is

$$dX_t = \sigma dW_t,$$

which implies $X_t = \sigma W_t$.

[2 marks]

Therefore by Feynman-Kac,

$$\begin{aligned}u(t, x) &= e^{-r(T-t)} \mathbb{E}[e^{\theta X_T} | X_t = x] \\ &= e^{-r(T-t)} \mathbb{E}[e^{\theta(x + \sigma(W_T - W_t))}] \\ &= e^{\theta x - r(T-t)} \mathbb{E}[e^{\theta \sigma(W_T - W_t)}].\end{aligned}$$

[3 marks]

Notice that $W_T - W_t \sim N(0, T - t)$, so we have

$$\begin{aligned}\mathbb{E}[e^{\theta \sigma(W_T - W_t)}] &= \int_{-\infty}^{\infty} e^{\theta \sigma x} \frac{1}{\sqrt{2\pi(T-t)}} e^{-\frac{x^2}{2(T-t)}} dx \\ &= e^{\frac{\theta^2 \sigma^2}{2}(T-t)} \frac{1}{\sqrt{2\pi(T-t)}} \int_{-\infty}^{\infty} e^{-\frac{(x - (T-t)\theta\sigma)^2}{2(T-t)}} dx \\ &= e^{\frac{\theta^2 \sigma^2}{2}(T-t)}\end{aligned}$$

[4 marks]

Therefore,

$$u(t, x) = e^{\theta x + (\frac{\theta^2 \sigma^2}{2} - r)(T-t)}.$$

[1 mark]

Comment

(a) This question is standard. Appeared in homework, though it was called Ornstein-Uhlenbeck process there.

(b) This question is slightly challenging as some students may forget how to find the value of the expectation and MGF for normal is not provided in the formula sheet.

CHECKING